



Product & methodology explainer

How it works, and **why the math** is the way it is.

A plain-English guide to the engine behind GiffMeMoney — why we use Markowitz, CAPM, Monte Carlo and the rest; the end-to-end flow from data to decision; how it beats investing by gut; and an honest account of what it still lacks to be a real trading system.

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03 The math & why we use each

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Why investing "by feel" usually loses.

Most individual investors underperform the market — not because the ideas are hard, but because humans are wired badly for it. GiffMeMoney exists to replace four predictable failure modes with a disciplined, transparent process.

Emotion & timing

People buy near the top (greed) and sell near the bottom (fear). A model has no emotions — it follows the same rules in a panic as in a boom.

Narrow view

A person watches a handful of stocks and one or two signals. The engine evaluates the whole universe through **73 strategies at once**, every refresh.

No sense of odds

"It'll probably go up" isn't a number. The engine outputs an **expected return, a range, and a probability** for five time horizons.

Ignored risk

Beginners chase return and forget the downside. Every recommendation here carries volatility, drawdown, Value-at-Risk and a bear-case scenario.

The goal isn't to be a fortune-teller — it's to make **consistent, evidence-based, risk-aware decisions** and to *show its work*, so you can trust (or challenge) every call.

No single indicator is right. So we vote.

Every famous strategy works *sometimes* and fails other times — momentum shines in trends and gets whipsawed in chop; value wins over years but lags for months. The central design choice of GiffMeMoney: treat each strategy as **one expert casting a vote**, then combine them intelligently.

1

Many experts

73 named, published strategies each score an asset from -100 (sell) to +100 (buy) with a confidence.

2

Weighted consensus

Votes are blended by confidence and reliability, then shrunk when the experts disagree — so it's only loud when they agree.

3

Honest forecast

The consensus becomes a composite score, a recommendation, and a five-horizon projection with a credible range.

This "ensemble" approach is exactly why it's more robust than any one tactic: the weaknesses of one method are offset by the strengths of others, and disagreement is treated as **uncertainty** rather than ignored. It's the same logic that makes a panel of forecasters beat any single forecaster.

Why "blend", not "pick the best"?

The best strategy *last year* is rarely the best next year. Blending avoids betting everything on a regime that may already be over.

Why confidence-weighting?

A signal screaming on rich data should count more than a weak signal on thin data. Confidence makes the vote meritocratic.

Every technique earns its place.

These aren't buzzwords — each is a Nobel-/industry-grade method chosen to answer a specific question. Here's what each does and **why it's in the engine**.

Markowitz Mean–Variance (MPT)

$$\max (\mu \cdot w - R_f) / \sqrt{w \cdot \Sigma \cdot w}$$

Why: finds the mix of assets with the most return per unit of risk — the "efficient frontier." It's **why we don't put everything in one pick**: diversification mathematically lowers risk without killing return. Powers the advisor & the bot's sleeves. (Markowitz, Nobel 1990.)

CAPM

$$E[R] = R_f + \text{beta} \cdot (E[R_m] - R_f)$$

Why: separates the return you deserve for taking *market* risk (beta) from genuine edge (alpha). It anchors our return forecasts to a sane baseline so nothing projects fantasy numbers. (Sharpe, Nobel 1990.)

Sharpe & Sortino Ratios

$$(\bar{R} - R_f) / \sigma$$

Why: a 30% gain with wild swings is worse than 20% smooth. These rank assets by **risk-adjusted** return so we reward quality, not luck. Sortino only punishes *downside* volatility.

Monte Carlo (GBM)

$$dS = \mu \cdot S \cdot dt + \sigma \cdot S \cdot dW$$

Why: the future isn't one number, it's a distribution. We simulate thousands of price paths to get the **probability of profit** and the realistic best/worst range — not a single guess.

GARCH(1,1)

$$\sigma_t^2 = \omega + \alpha \cdot \varepsilon_t^2 + \beta \cdot \sigma_{t-1}^2$$

Why: volatility comes in clusters (calm, then storms). GARCH forecasts *tomorrow's* risk from recent shocks, so position sizes and bands react to changing conditions.

Value-at-Risk / CVaR

$$\text{CVaR} = E[\text{loss} \mid \text{loss} \geq \text{VaR}]$$

Why: answers "how bad is a bad day?" CVaR (expected shortfall) sizes the **tail risk** so the app surfaces downside instead of hiding it.

From cheapness to momentum to quality.

Fama–French Factors

$$R - R_f = \alpha + b \cdot \text{Mkt} + s \cdot \text{SMB} + h \cdot \text{HML}$$

Why: decades of data show size & value factors explain returns beyond the market. We use them to judge whether an asset's return is **cheap exposure or real skill**. (Fama, Nobel 2013.)

Value — Graham & Magic Formula

$$\text{EarnYield} = \text{EBIT}/\text{EV} ; \text{ROC} = \text{EBIT}/\text{Capital}$$

Why: buying good businesses cheaply is the oldest edge that works. Greenblatt's formula ranks cheap-and-profitable; Graham's rules add a margin of safety.

Momentum & Trend

$$12\text{-}1 \text{ return} ; \text{price vs } 200\text{-day MA}$$

Why: winners tend to keep winning over 3–12 months (one of the most robust anomalies in finance). Trend rules also tell the bot **when to step aside** in downtrends.

Mean-Reversion

$$z = (\text{price} - \mu) / \text{sigma} ; \text{RSI, Bollinger}$$

Why: short-term overreactions snap back. These capture the opposite edge to momentum, so the ensemble works in both trending and choppy markets.

Black–Scholes

$$C = S \cdot N(d_1) - K \cdot e^{(-rT)} \cdot N(d_2)$$

Why: the language of options & implied risk; lets us reason about leverage and the market's priced-in uncertainty. (Merton/Scholes, Nobel 1997.)

Backtesting engine

$$\text{CAGR} \cdot \text{Sharpe} \cdot \text{MaxDD} \cdot \text{win-rate}$$

Why: a score means nothing without a track record. We replay each timing strategy bar-by-bar over history vs buy-and-hold, so claims are **proven, not asserted**.

The throughline: value tells us *what's cheap*, quality & factors tell us *what's good*, momentum/trend tell us *what's working now*, and Markowitz/CAPM/Monte-Carlo/VaR tell us *how to size it and what can go wrong*. Together they cover the whole decision.

How it turns votes into a credible forecast.

A naïve model would compound a daily guess and print "+800% in 5 years." We deliberately built the projection layer to be **honest**, in five steps:

1 Ensemble the drift

Combine the projecting strategies' expected daily returns, weighted by confidence.

2 Shrink toward sanity

Pull that estimate toward a CAPM baseline (James–Stein shrinkage) and **cap it** to realistic ranges — this is what kills the fantasy numbers.

3 Read the regime

Detect bull/bear & calm/stormy from trend + volatility, and tilt the forecast accordingly.

4 Forecast the range, not a point

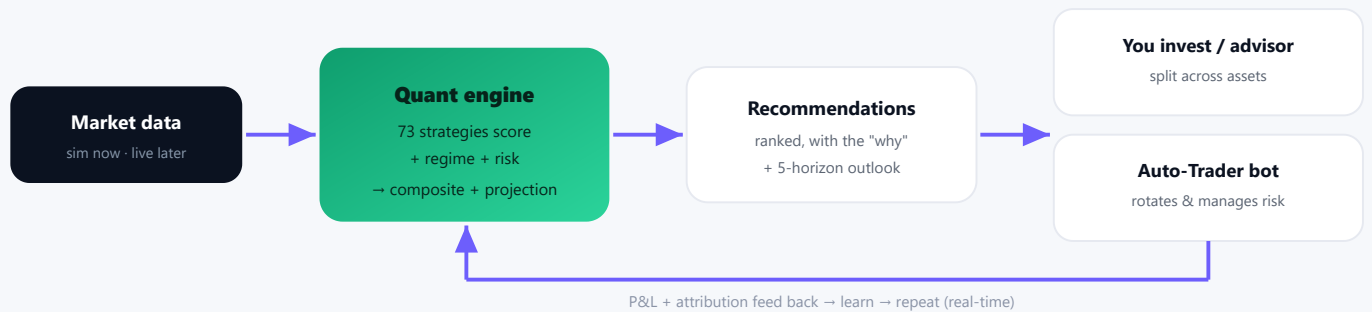
Use GARCH volatility + fat-tailed/bootstrapped bands for credible 5th–95th-percentile outcomes, plus a bull/base/bear scenario fan and the probability of a gain.

5 Stay consistent

The same engine powers the projection *and* the Monte-Carlo, so the two always agree — no contradictory numbers.

This is why a real run showed Bitcoin at a believable **+173% over 5 years** (not +800%), with the downside (a -63% bear year) shown right next to it. Honest beats impressive.

From raw market data to your decision.



1 · Analyze

Prices stream in; the engine re-scores the universe and updates projections continuously.

2 · Decide

You see ranked ideas with reasons, or let the advisor build an optimal basket for your risk level.

3 · Act & track

Invest (or let the bot run), then watch real-time P&L and see exactly which strategy helped or hurt.

GiffMeMoney vs. investing by gut.

DIMENSION	MANUAL INVESTING	GIFFMEMONEY
Coverage	A few names, 1–2 signals	Whole universe × 73 strategies , every refresh
Discipline	Mood-driven; inconsistent	Same rules every time, no emotion
Odds & horizon	"It'll probably go up"	Expected return + range + P(gain) over 1D→5Y
Risk awareness	Usually an afterthought	Beta, vol, VaR, drawdown, bear case on every call
Evidence	Hunches & headlines	Backtested track record vs buy-and-hold
Diversification	Guessed	Markowitz-optimized allocations
Speed	Hours of manual research	Seconds; continuous
Transparency	—	Shows the formula & reason behind every score

The honest framing: it doesn't promise to beat the market — it removes the dumb mistakes (no diversification, no risk control, emotional timing, no track record) that cause most individual investors to *underperform* it. That alone is a large, realistic edge.

A bot that manages money like a desk, not a gambler.

The Auto-Trader runs the whole pipeline automatically across five selectable modes — and it's built to be **honest about losses**.

Splits capital into sleeves

Spreads the budget across the best assets per the chosen objective (max-Sharpe, min-variance, momentum, risk-parity).

Analyzes before every rebalance

Re-reads the market regime + the 73-signal composite, then re-allocates — roughly monthly, not frantically.

Rotates toward what works

Shifts capital to recent winners and **away from losers** (bandit-style). Crucially, it is **anti-martingale**: it never doubles down on a loser to "win it back" — the way accounts blow up.

Controls risk

Per-sleeve stop-loss, a portfolio drawdown circuit-breaker that raises cash, and volatility-aware sizing.

Tells you the truth

Every run reports the equity curve **vs buy-and-hold**, plus which strategy did best and which did **worst** — so you can judge it, not just trust it.

Five options to compare

Conservative · Balanced · Aggressive · Adaptive-Bandit · All-Weather — run side by side to see which fits your risk.

In testing it **underperformed simple buy-and-hold in some runs** — and the app shows that plainly. That's the integrity of the tool: it reveals when "smart" trading would have lost to doing nothing.

Straight answers.

Is this trading real money right now?

No. It runs on a realistic **simulator** with a paper wallet — nothing real moves. Live data and a real broker (paper-first) are built in and switch on with your own keys.

Does it guarantee profit?

No, and anything that claims to is lying. Markets are uncertain. It improves your *process* and odds; it does not remove risk. You can lose money.

Where does the data come from?

Today: a deterministic market simulator (no keys needed). Tomorrow: real feeds — Finnhub/Polygon for stocks, CoinGecko/Binance for crypto — behind the same interface.

How is it "AI"?

It applies 73 quantitative models + regime detection to analyze every asset before acting. It's transparent, explainable math — not a black box.

How accurate are the projections?

They're **calibrated estimates with ranges**, deliberately conservative. They describe a distribution of outcomes, not a promise. On synthetic data they're illustrative; real accuracy requires live-data validation.

Is my money / data safe?

Passwords are hashed (PBKDF2), sessions use signed tokens, and no real funds are handled in the current build. Card entry in the sandbox is masked and never charged.

Can I use it for myself?

Yes — it's designed for personal use. Run it locally or via Docker; connect real data in paper mode and validate before risking anything.

What it doesn't have (yet).

A credible product owns its limits. Here's what's missing on the feature side.

CAPABILITY	STATUS	NOTE
Live market data wired by default	No	Built & pluggable — needs your API keys
Real brokerage execution (live orders)	No	Paper works; live is gated OFF by design
News / sentiment / social signals	No	Engine is price + fundamentals + factors only
Machine-learning / alpha models	No	Classical quant only (intentionally explainable)
Options / futures / margin trading	No	Black-Scholes is analytic only, not executed
Tax lots, fees, slippage modeling	Basic	Simple cost model; no tax/accounting
Mobile app / push alerts	No	Responsive web only
Cloud hosting / multi-device sync	No	Runs locally / self-hosted (Docker provided)
Real fundamentals & factor feeds	Partial	Approximated where live feeds don't provide them

These are **additions, not flaws** — the core engine is complete and tested (362 automated tests). Each item above is a known, scoped piece of future work.

What's left to make money — honestly.

Two parts: the **achievable engineering**, and the **part no software can fix**.

● A · ENGINEERING (DOABLE, MOSTLY YOUR KEYS)

1 · Connect live data

Add free Finnhub/CoinGecko keys so it analyzes real markets.

2 · Connect a broker (paper)

Alpaca paper keys → real prices, simulated fills, zero risk.

3 · Validate for weeks

Run live-but-paper and watch the *real* track record before any capital.

4 · Go live (your call)

Flip the gated live switch with your own funded account & keys.

● B · THE PART NO APP CAN DELIVER

A "100% money-making machine" does not exist — for anyone. Not for hedge funds, not for this app. Markets are competitive and uncertain; guaranteed profit is mathematically impossible. What you have is a **professional-grade research, risk-management and validation platform** — it tilts the odds and enforces discipline, but it cannot remove risk. The realistic plan: validate on live-paper, start tiny, only risk money you can afford to lose, and let the bot's *own* honest track record decide.



Educational simulation · not financial advice · capital is at risk